



X·EasyGo:An interactive journey bridging reality and digital storytelling through location-based AR technology

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Portfolio



Demo

Abstract

Traditional campus tours often rely on **one-way explanations**, making visitors **passive listeners** rather than **active explorers**. At Xi'an Jiaotong-Liverpool University (XJTLU), this is especially limiting because the university's **student-centered philosophy** is better understood through **movement, interaction, and experience**. Although existing academic and commercial solutions have explored location-based AR[1][2], campus navigation[3], and playful interaction[4], they often fail to combine **practical guidance, meaningful place-based learning, and user engagement** in one coherent experience. In response, this project proposes **X·EasyGo**, an **AR-supported campus tour system** designed for both **students and visitors**. It aims to provide **clear route-first navigation, role-specific landmark content, and lightweight interactive features** to make campus exploration more intuitive, engaging, and reflective of XJTLU's educational identity.



Requirement Gathering

Have Participants Used a Campus Tour Guide Before?

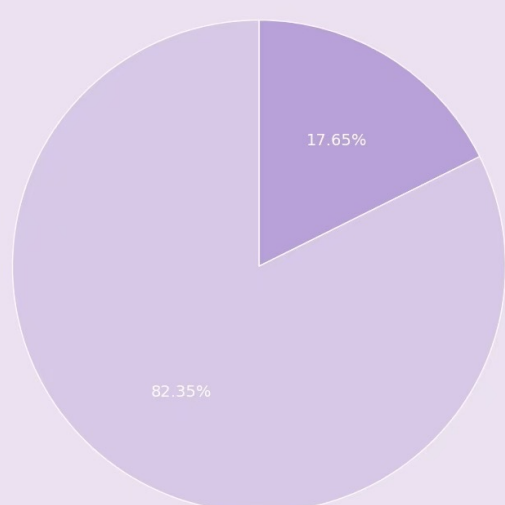


Fig 1. Prior Usage of Campus Tours

Attitude Survey Results

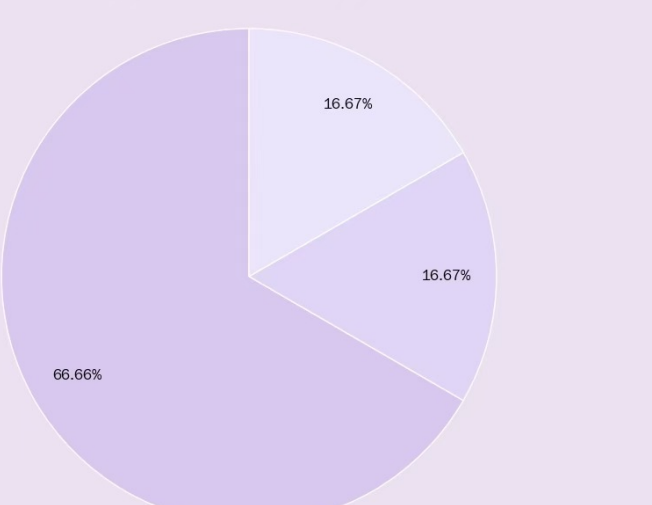


Fig 2. Attitude to the new tour product

Preferred Campus Tour Guide Format

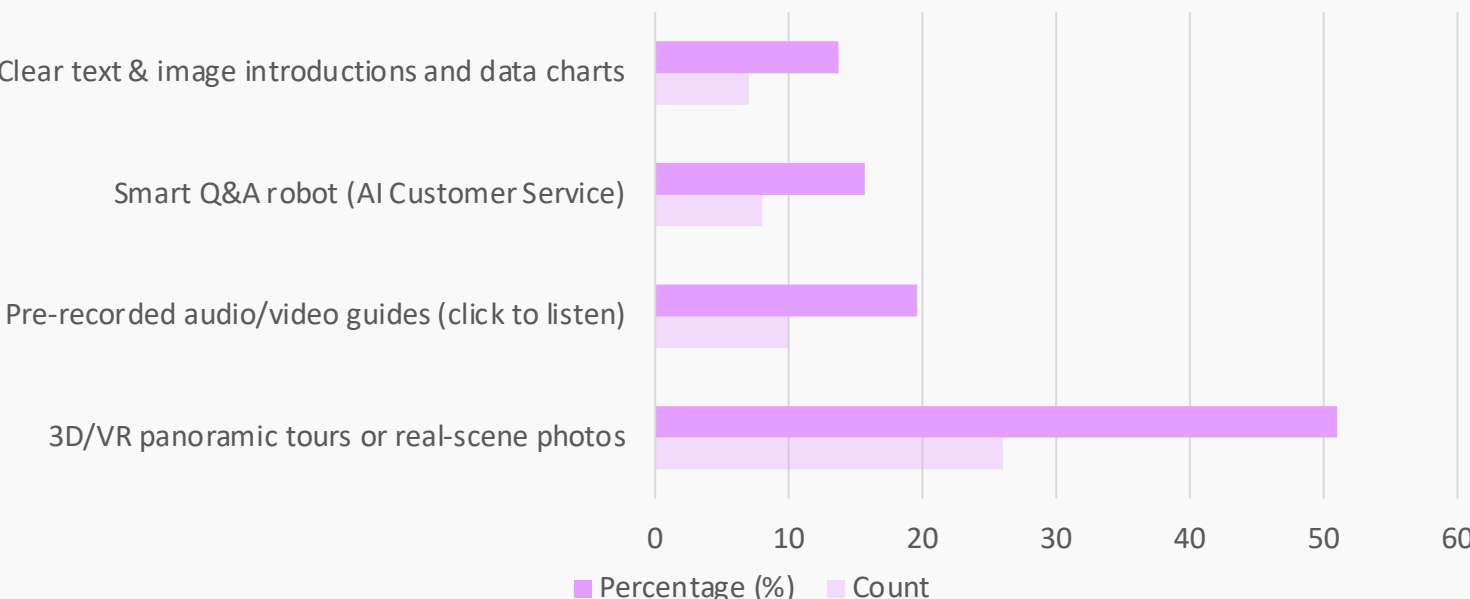


Fig 3. User Preferences for Campus Tour Guide Formats

User Pain Points	Core App Solutions
No customized content for visitors	• Role-specific explanations
Low interaction & engagement	• Gamified travel quests • AR-based guidance
No user-to-user connection	• Message board

Fig 4. Must-to-have functions and details

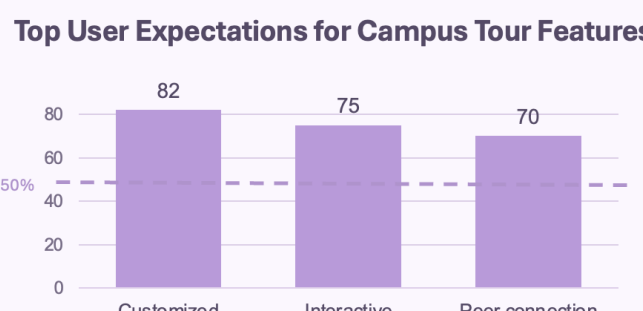


Fig 5. Functionality Recognition

Our investigation (n = 51) reveals that **82.35%** of participants have never used existing campus tour services (Fig. 1), while approximately **66.66%** expressed a willingness to use such a product to assist them in exploring the campus (Fig. 2). Furthermore, the survey indicates that **most participants prefer VR panoramic tours** as their preferred campus tour format (Fig. 3). In addition, participants expressed a strong expectation for **role-specific content, enhanced interactive engagement, and opportunities** for peer connection (Fig. 5).

Then we make the **persona** shown in Figure 6.

Lina Zhao
First-time Visitor

SCENARIO
During a short campus visit, Lina uses the mobile web app to explore the most representative places at XJTLU. She hopes the tour can be more than simple navigation, helping her quickly understand the university's atmosphere, academic character, and memorable landmarks through interactive features such as AR guidance and stamp collection.

Goals & Needs

- Discover the most iconic and meaningful places on campus efficiently
- Understand what different buildings represent, not just their names
- Experience the campus in a more engaging way than reading static signs
- Leave with a strong overall impression of XJTLU in a short amount of time

Pain Points

- Campus maps usually show locations, but not why places matter
- Traditional tours are often passive and easy to forget
- Building abbreviations and unfamiliar names are confusing
- Short visits make it hard to form a clear understanding of the campus

Ethan Wu
Regular User

SCENARIO
In the first two weeks of the semester, Ethan uses the mobile web app while walking between classes and service points. He wants to quickly find useful places such as study areas, service desks, and academic buildings, while also learning how the campus is organised. He prefers a guide that feels practical, lightweight, and motivating rather than a long formal introduction.

Goals & Needs

- Find important places for study, support, and daily campus life quickly
- Understand the practical functions of different buildings
- Learn practical information for studying, services, and campus routines
- Feel less confused when moving around campus for the first time

Pain Points

- The campus scale feels large and unfamiliar
- Building abbreviations are difficult to understand at first
- Existing maps do not explain what spaces are actually useful for
- It is hard to quickly connect places with real student needs

Fig 6. Persona

The **user journey map** in Figure 7 illustrates how users interact with our system to address key pain points.

Visitor Journey

	Before Visit	Arrival	Exploration	Understanding	Moving On	After Visit	Feedback
Action	Gets a tour plan.	Enters campus and looks for buildings.	Walks between stops and listens.	Hears introductions about key spaces.	Moves to the next stop.	Leaves with a general impression.	Leaves a short comment or rating.
Pain Point	Knows little about XJTLU.	Layout and abbreviations are confusing.	The visit feels passive and one-way.	Campus philosophy feels abstract.	There is little continuity.	Remembers places, not meaning.	No easy way to record impressions right away.
Feeling	Curious, unsure.	Slightly confused.	Interest starts to drop.	Hard to connect.	Tired, detached.	Impression is incomplete.	Wants to share the visit.
Opportunity	Show a quick campus overview.	Use clear maps and landmark names.	Add scan check-ins, stamp collection, and virtual study guidance.	Explain how each space reflects student-centred learning.	Add progress cues and next-stop prompts.	Summarise key spaces and values.	Add quick comments, ratings, and reflection notes.

Student Journey

	Before Visit	Arrival	Exploration	Understanding	Moving On	After Visit	Feedback
Action	Knows names, not functions.	Tries to find useful places.	Follows a campus route.	Learns about a building.	Goes to the next place.	Finishes the route.	Leaves tips or saves useful notes.
Pain Point	Useful info is scattered.	Cannot tell where to study or get help.	Traditional tours ignore daily student needs.	The explanation is too general.	There is no sense of progress.	Still may not know where to go later.	Useful impressions may be forgotten later.
Feeling	Needs guidance.	Overwhelmed.	Only partly engaged.	Not sure it helps.	Loses momentum.	Partly informed.	Wants practical takeaways.
Opportunity	Show practical campus info first.	Label spaces by purpose.	Add scan check-ins, stamp collection, and virtual study guidance.	Show quiet floors, facilities, and services.	Use check-ins, rewards, and route progress.	Reinforce memory with summaries and saved spots.	Let users save tips, comments, and favourite spots.

Fig 7. User Journey Map



Prototype Design & Iterations

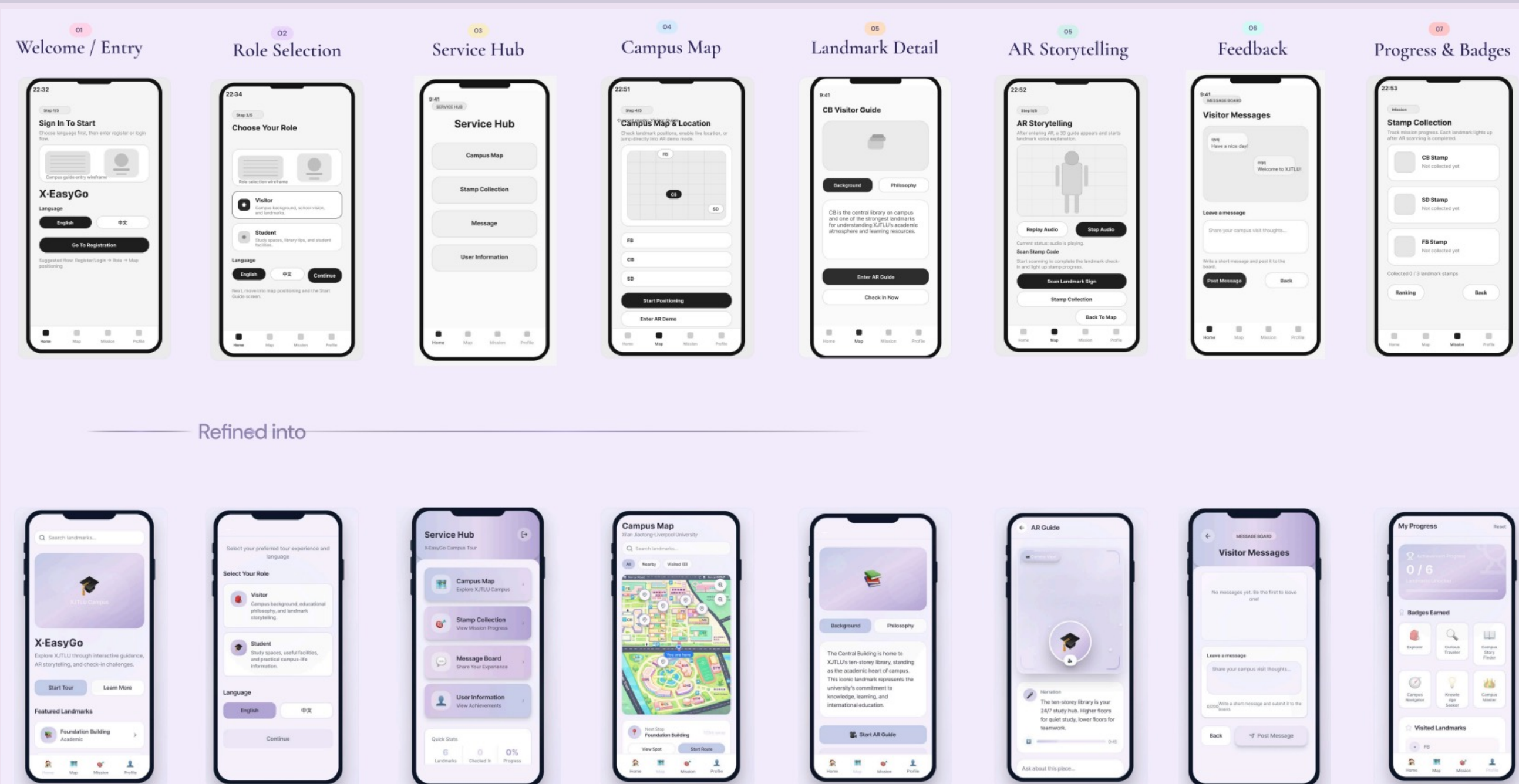


Fig 8. Iterations of app prototype UI design

System Architecture

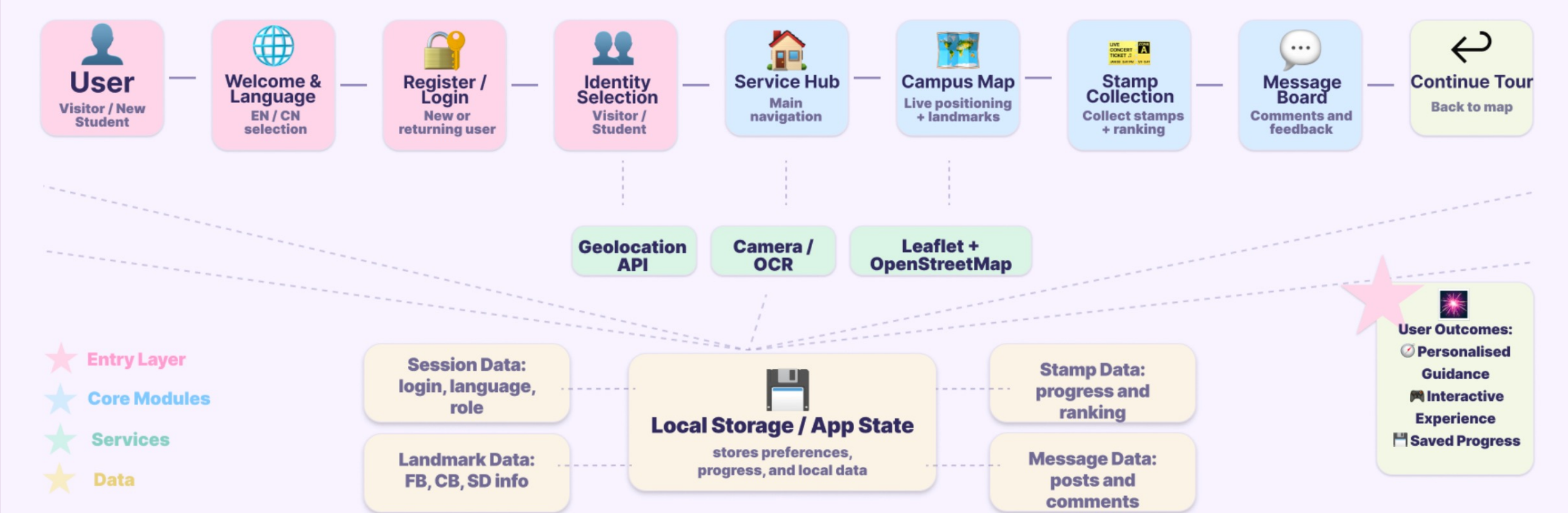


Fig 9. System Architecture

Figure 8 illustrates the iterative design process of the **app prototype UI**. We first developed a basic version of the system and then progressively added more features and a more **visually appealing interface**. Figure 9 shows the **system workflow**, illustrating how user interactions move through **core modules, supporting services, and data storage**.

To balance low-friction navigation with playful interaction, we explored **two alternative stamp collection mechanisms**:
(1) an automatic location-based trigger, focusing on efficiency.
(2) a manual AR scanning interaction, emphasizing user engagement.
This design dilemma informed the **development of two prototype variants**, which were later evaluated through user testing.



Evaluation Results

In the evaluation, we invited **10 participants** to use and experience our product.

A/B Testing

Participants (n = 10) were divided into two groups to conduct an A/B test comparing two approaches to stamp collection:

- Prototype A:** Users automatically receive a stamp when the location system detects that they are near a building.
- Prototype B:** Users must scan the signboard outside the building to obtain a stamp.

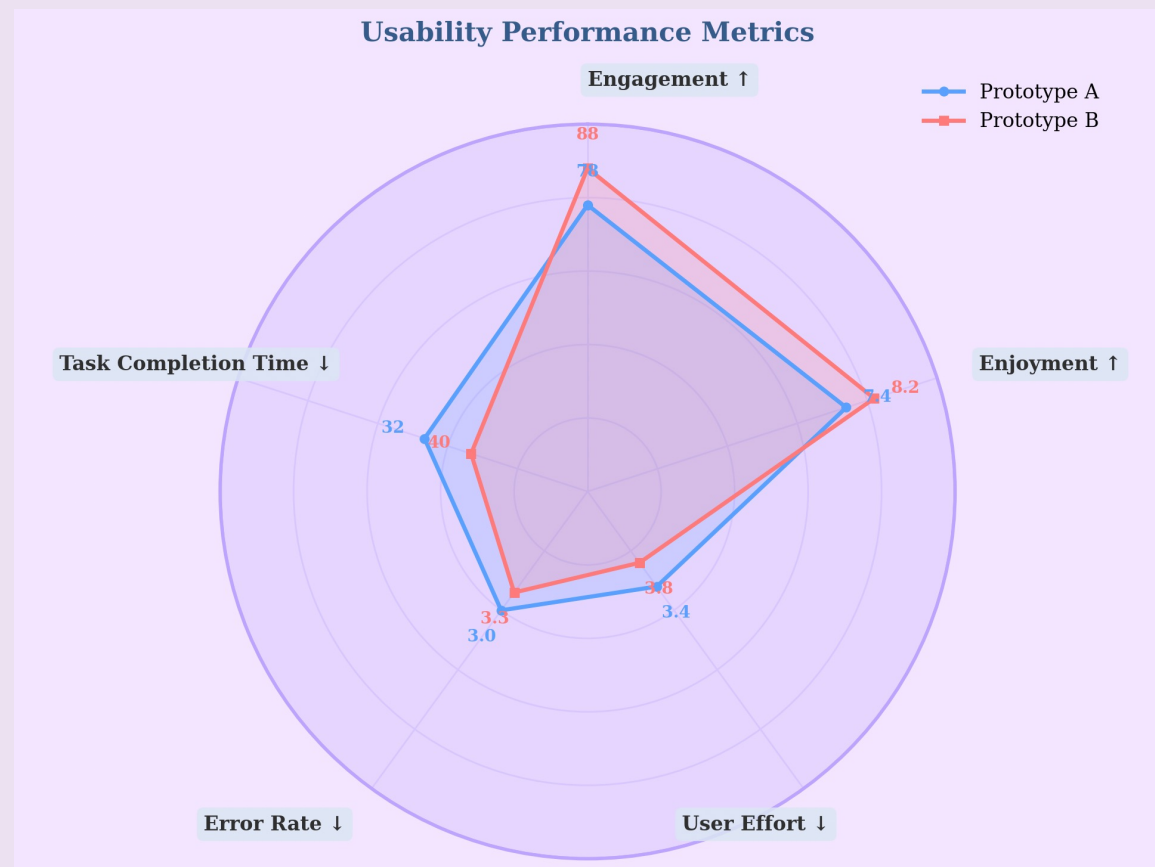


Fig 10. Usability Performance Metric

As shown in Figure 10, **Prototype A** achieves **higher scores in task completion time, error rate, and user effort**. However, **Prototype B** performs better in **engagement and enjoyment**, which are key factors in the **user requirements**. Therefore, despite its drawbacks, **Prototype B provides greater overall value to the user experience**. As a result, Prototype B is selected as the final design.



Conclusion

Results suggest that X·EasyGo **improves campus exploration** by combining **clear guidance, role-specific content, and interactive AR features** in a more **engaging and user-centred way**.

Limitations

- The AR system is still at a **preliminary stage**, with limited positioning accuracy and delays in the scan-and-collect-stamp function.
- The current prototype covers **only three campus buildings**, which limits the scope and completeness of the experience.
- The current evaluation is **based on a limited number of participants**, so broader user testing is still needed.

Future Work

- Expand the system to **cover more campus buildings and functional areas**.
- Improve the underlying technology** to enhance **positioning accuracy and reduce response delays**.
- Conduct broader testing with different user groups** to further refine interaction design and content delivery.



References & Acknowledgments

- [1] A. Klefodimos and A. Evangelou, "Location-based augmented reality in education," *Encyclopedia*, 2025, 5, 54, doi: 10.3390/encyclopedia5020054.
- [2] X. Fonseca, P. Spangenberg, M. Baer, R. Schmidt, and H. Söbke, "Location-based augmented reality in education: A systematic literature review," *Computers and Education Open*, 9 (2025) 100277, doi: 10.1016/j.cao.2025.100277.
- [3] S. Tasyonar and A. M. Uzun, "Using a location-based augmented reality mobile application to support the new students' orientation process," *Open Praxis*, 17(4), pp. 768–784, 2025, doi: 10.55982/openpraxis.17.4.886.
- [4] C.-Y. Hsieh, C.-M. Chen, and Y.-C. Yang, "A game-based augmented reality navigation system to support makerspace user education in a university library," in *2023 14th IIAI International Congress on Advanced Applied Informatics (IIAI-AAI)*, 2023, pp. 218–223, doi: 10.1109/IIAI-AAI59060.2023.00052.